JAVA WITH DSA-DAY 6

2 POINTERS

REVERSE VOWELS IN A STRING

```
import java.util.*;
class Main {
    static boolean isVowel(char c) {
        c = Character.toLowerCase(c);
        return c == 'a' || c == 'e' ||
               c == 'i' || c == 'o' || c == 'u';
    }
    public static void main(String[] args) {
        String s = "leetcode";
        char[] arr = s.toCharArray();
        int left = 0;
        int right = arr.length - 1;
        while (left < right) {
            while (left < right && !isVowel(arr[left])) {
                left++;
            }
            while (left < right && !isVowel(arr[right])) {
                right--;
            }
            char temp = arr[left];
            arr[left] = arr[right];
            arr[right] = temp;
            left++;
            right--;
        }
        System.out.println(new String(arr));
    }
}
```

OUTPUT:

IceCreAm

AceCreIm

2SUM PROBLEM:

(2 pointer not suitable for unsorted array)

```
class Main{
  public static void main(String[] args) {
    int arr[] = {2,7,11,15};
    int left = 0;
    int right = arr.length-1;
    int sum = 0;
    int target = 9;
    while (left < right) {
      sum = arr[left]+arr[right];
      if (sum == target){
        System.out.println (left+" "+right);
        return;
      }
      else if (sum < target) {
        left++;
      }
      else {
        right--;
      }
    }
    System.out.println ("-1 -1");
  }
}
```

OUTPUT:

0 1

REMOVE THE DUPLICATES IN AN ARRAY:

```
class Main {

    public static void main(String[] args) {

        int arr[]={1,1,2,3,3,4};

        int left=0;

        for(int right=1;right<arr.length;right++){

            if(arr[left]!=arr[right]){

                left++;

                arr[left]=arr[right];

            }
```

```
        }
        for(int i=0;i<=left;i++){
            System.out.print(arr[i]+" ");
        }
    }
}
```

OUTPUT:

1 2 3 4

Using while

```
class Main {
    public static void main(String[] args) {
        int arr[] = {1,1,2,3,3,4};
        int left = 0;
        int right = 1;
        while (right < arr.length) {
            if (arr[left] != arr[right]) {
                left++;
                arr[left] = arr[right];
            }
            right++;
        }
        for (int i = 0; i <= left; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

OUTPUT:

1 2 3 4

2 POINTERS IN STRING:

PALINDROME OR NOT:

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the string: ");
        String s = sc.nextLine();
        String st = s.toLowerCase();
        int left = 0;
        int right = st.length() - 1;
        while (left < right) {
            if (st.charAt(left) != st.charAt(right)) {
                System.out.print(s + " is not a palindrome");
                return;
            }
```

```
            left++;
            right--;
        }
        System.out.print(s + " is a palindrome");
    }
}
```

OUTPUT:

Enter the string

Madam

Madam is a palindrome

REMOVE 1 ELEMENT FROM THE STRING AND IF THEN IT IS A PLAINDROME, PRINT TRUE OR FALSE

Example:
Madabm

b

Is a palindrome

Madabbm

b

b

Is not a palindrome

```
import java.util.*;

class Main {

    static boolean check(String s, int left, int right) {

        while (left < right) {

            if (s.charAt(left) != s.charAt(right)) {

                return false;

            }

            left++;

            right--;

        }

        return true;

    }

    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the string:");
        String s = sc.nextLine().toLowerCase();
        int left = 0;
        int right = s.length() - 1;
        while (left < right) {
            if (s.charAt(left) == s.charAt(right)) {
                left++;
                right--;
            }
            else {
                if (check(s, left + 1, right)) {
                    System.out.println(s.charAt(left));
                    System.out.println("Is a palindrome");
                    return;
                }
                else if (check(s, left, right - 1)) {
                    System.out.println(s.charAt(right));
                    System.out.println("Is a palindrome");
                    return;
                }
                else {
                    System.out.println(s.charAt(left));
                    System.out.println(s.charAt(right));
                    System.out.println("Is not a palindrome");
                    return;
                }
            }
        }
        System.out.println("Is a palindrome");
    }
}
```